Final Project Report

COVID-19's Impact on Motor Vehicle Collisions in NYC

Group 1 (STAT 605 - S1)

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1 Introduction

The COVID-19 pandemic created an unprecedented natural experiment in urban mobility. Beginning March 2020, New York City's shelter-in-place orders fundamentally altered how millions of people moved through one of the world's densest transportation networks. This project analyzes how COVID-19 transformed motor vehicle collision patterns across frequency, severity, temporal distribution, and geography.

Using 13 years of police-reported crash data (2012-2025), we examine three distinct periods: Pre-COVID (baseline), COVID (March 2020-December 2021), and Post-COVID (2022-present). Our analysis addresses four questions: (1) Did collision rates return to pre-pandemic levels or establish a "new normal"? (2) Did crashes become more or less deadly during reduced traffic? (3) Were impacts uniform across NYC's five boroughs? (4) How did vehicle characteristics and driver behaviors contribute to collision severity across periods?

2 Data

2.1 Primary Dataset: NYC Motor Vehicle Collisions

Source: NYC OpenData (2.22M records, 29 variables, 2012-2025)

Police-reported collisions from NYPD's TrafficStat system. Key variables include crash date/time, location (coordinates, borough), casualties (injuries/fatalities by road user type), contributing factors (up to 5 per crash), and vehicle types (up to 5 per crash).

2.2 Secondary Dataset: Vehicle-Level Details

Source: NYC OpenData (4.45M records, 25 variables, 2016-2025)

Vehicle-specific data linked via `COLLISION_ID`. Each row represents one vehicle in a crash. Variables include vehicle characteristics (make, model, year, type, occupants), driver demographics (sex, license status), pre-crash actions, damage locations, and contributing factors. We aggregated to crash-level metrics: vehicle count per collision, average vehicle age, and average occupants per vehicle.

2.3 Tertiary Dataset: NYC Speed Limits

Source: NYC OpenData (141K street segments, Vision Zero initiative)

Posted speed limits by street segment with geographic coordinates. We matched segments to crash locations to analyze whether high-speed roads (¿40 mph) correlate with crash frequency and severity changes during COVID. NYC's default speed dropped from 30 to 25 mph in 2014; higher limits indicate highways and major roads.

2.4 Quaternary Dataset: Twitter Traffic Reports

Source: Research dataset by Dabiri (2018); 51,178 classified tweets

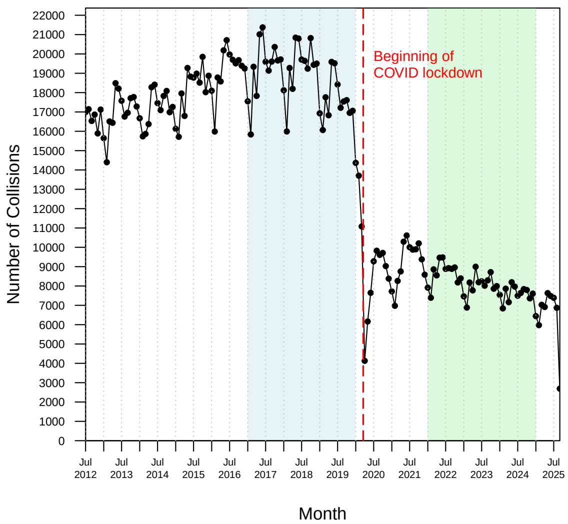
Tweets categorized as non-traffic (0), traffic incidents (1), or traffic conditions (2). We filtered to 798 NYC-specific tweets, then isolated 66 vehicle crash reports for text mining. While predating COVID, this establishes a baseline for social media reporting patterns and enables contributing factor text analysis from official crash reports.

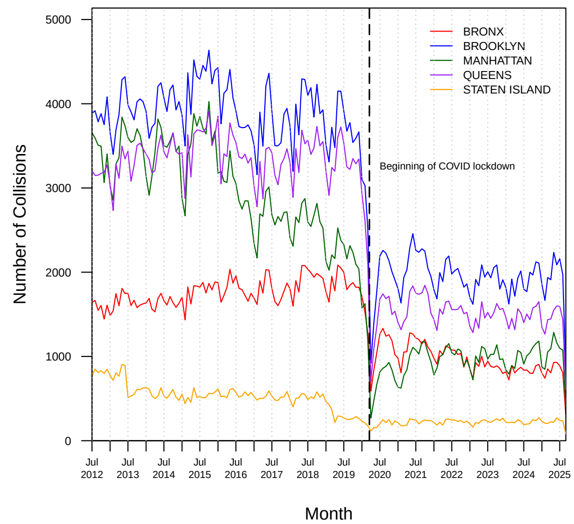
3 Exploratory Data Analysis

3.1 Did collision rates return to pre-pandemic levels or establish a "new normal"?

3.1.1 Motor Vehicle Collisions over Time

We begin by examining how the overall number of vehicle collisions in NYC evolved over time. Monthly crash counts show a dramatic decline in early 2020, coinciding with the onset of COVID-19 restrictions and the citywide lockdown. This sharp drop reflects the sudden reduction in traffic volume as commuting, tourism, and social activity came to a halt.





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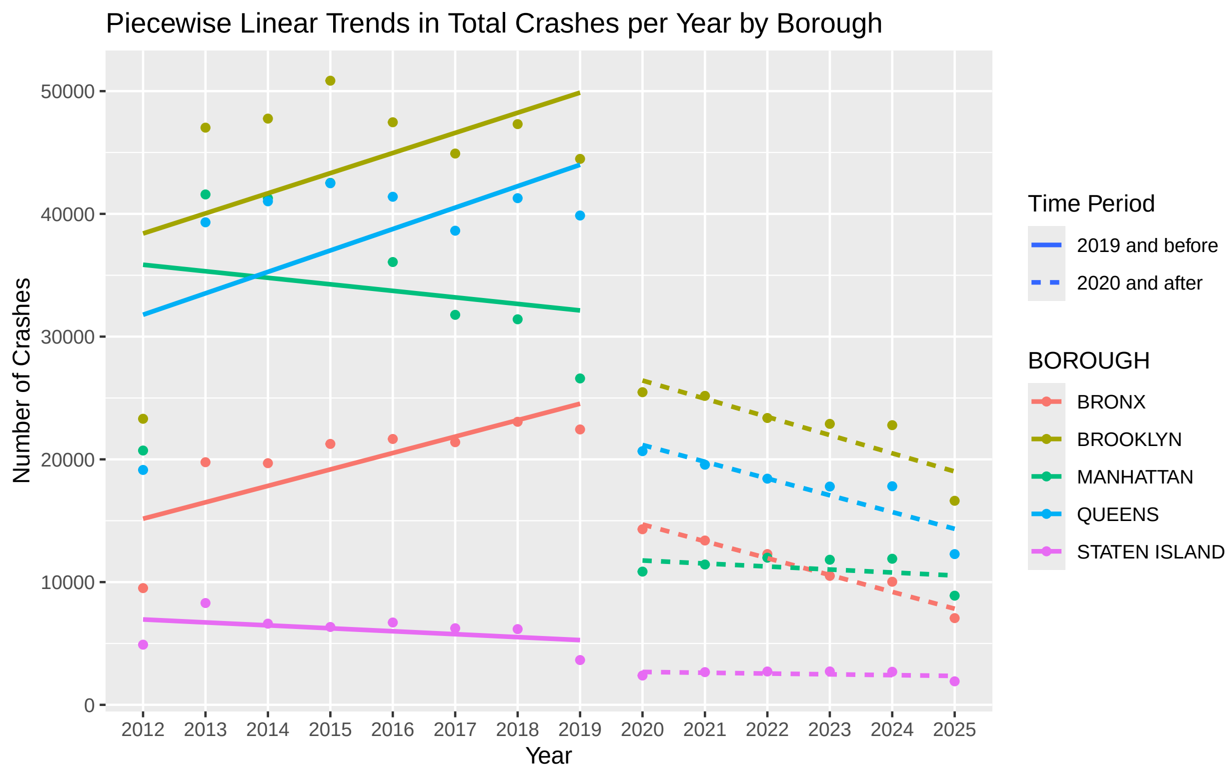
When broken down by borough, the same general pattern appears, though the magnitude of change differs. Brooklyn and Queens experienced the largest declines and slowest recoveries, consistent with their heavy commuter traffic. Manhattan saw a steep initial drop followed by a quicker rebound, reflecting its concentration of essential services and commercial traffic. Staten Island and the Bronx showed smaller relative changes, indicating more stable local travel patterns.

Overall, these trends reveal that while traffic volumes rebounded after the lockdown, the city's collision rates did not fully recover. The data suggest a structural shift in how, when, and why people travel in NYC.

3.1.2 Yearly Crash Trends by Borough, 2012–2025

From the earlier graphs, it seemed like there may be a difference in the trend of the amount of collisions before and after COVID. Therefore to further analyze this, we performed a linear regression by borough of the total crashes per year, performing separate linear regressions before and after the COVID shutdown.



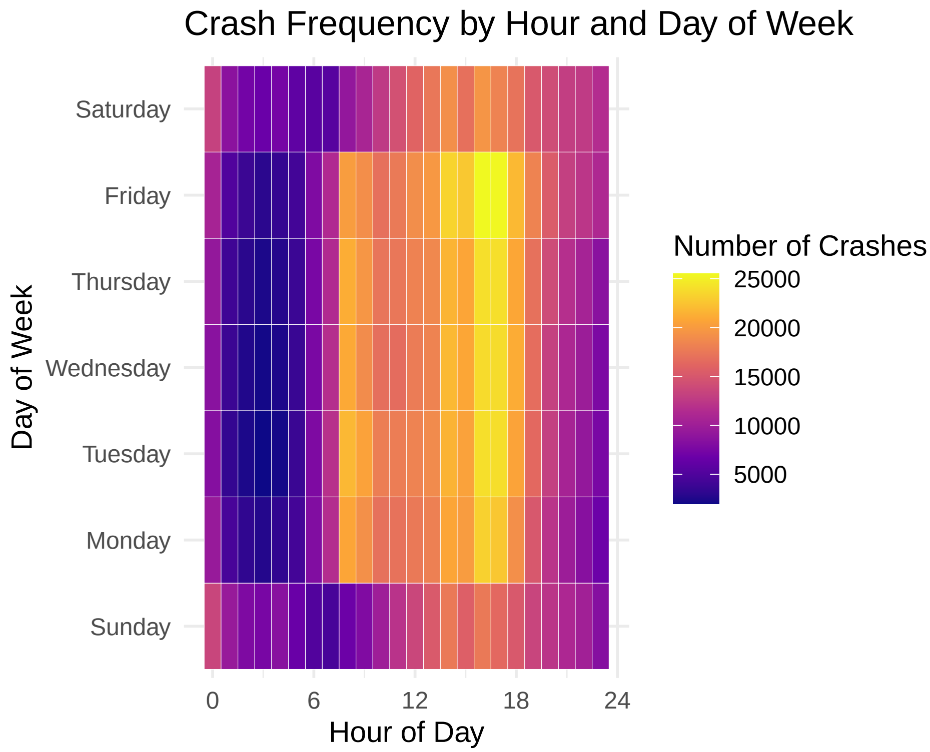
When broken down by borough, the same general pattern appears, though the magnitude of change differs. The piecewise linear trends shows yearly crash totals by borough from 2012 to 2025. Brooklyn and Queens experienced the largest declines and slowest recoveries, consistent with their heavy commuter traffic. Manhattan saw a steep initial drop followed by a slower rebound, reflecting lasting reductions in commuting and tourism. The Bronx showed a moderate decline followed by gradual recovery, while Staten Island remained relatively stable.

These results show that while the overall pattern of decline and partial recovery was shared citywide, the impacts were not uniform. Differences in population density, travel purpose, and road network composition likely explain why some boroughs experienced larger and more persistent changes than others.

3.2 When do crashes happen? Daily/weekly patterns

3.2.1 Crash Frequency by Hour and Day of Week

To understand when crashes are most likely to occur, we created a heatmap of the crash frequency by hour of the day and day of the week. Each tile represents the total number of crashes in that time slot, with brighter colors indicating more frequent crashes.

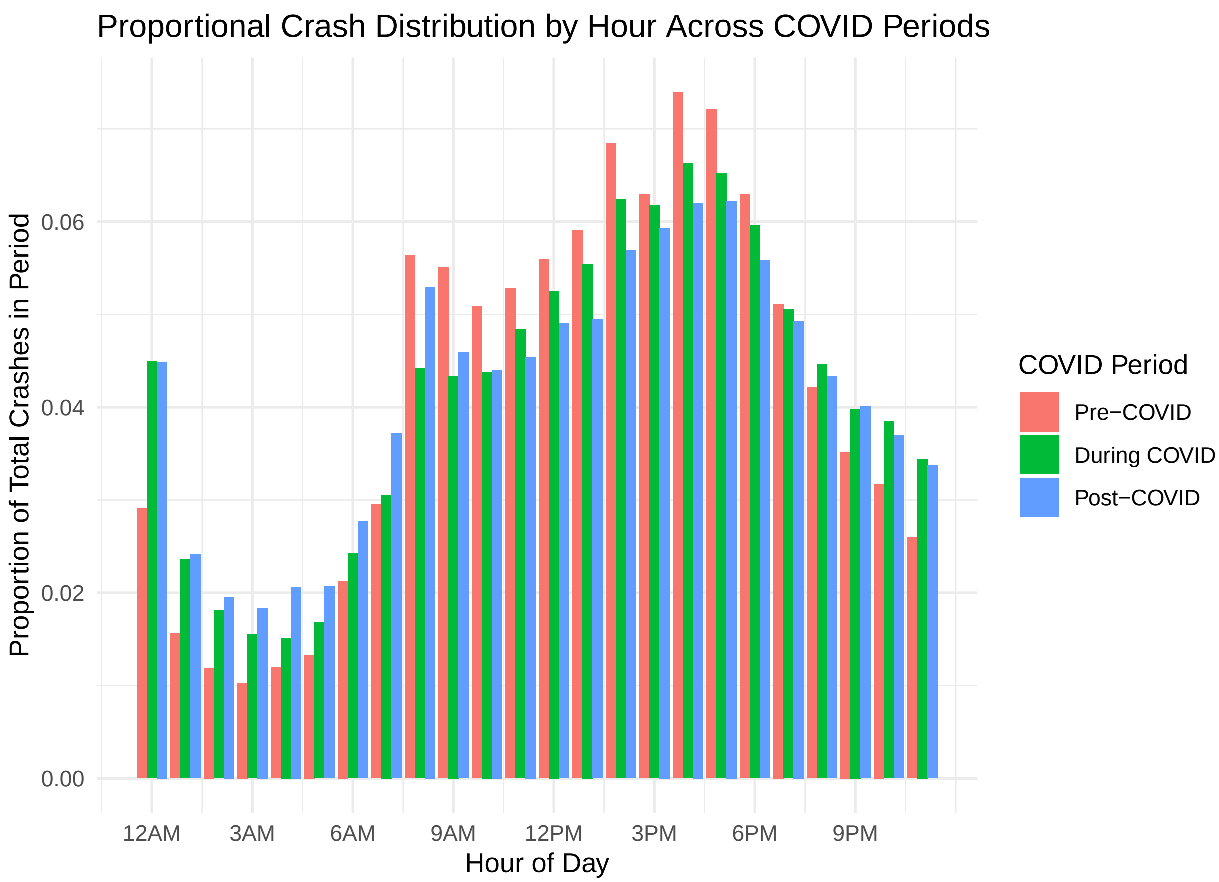


The heatmap shows a clear concentration of crashes during daytime and evening hours, especially between 8 AM and 7 PM across all weekdays. Friday stands out with the highest crash intensity, particularly in the late afternoon and early evening, likely reflecting a mix of rush-hour traffic and increased social activity at the end of the work week.

In contrast, the overnight hours (midnight to 6 AM) consistently display the fewest crashes across all days, reflecting lower traffic volumes during those times. Sundays appear somewhat less intense overall compared to weekdays, with a broader distribution of crashes throughout the afternoon.

Overall, the heatmap highlights how collision risk peaks during high-traffic commuting periods and social activity times, with Friday evenings being particularly hazardous in New York City.

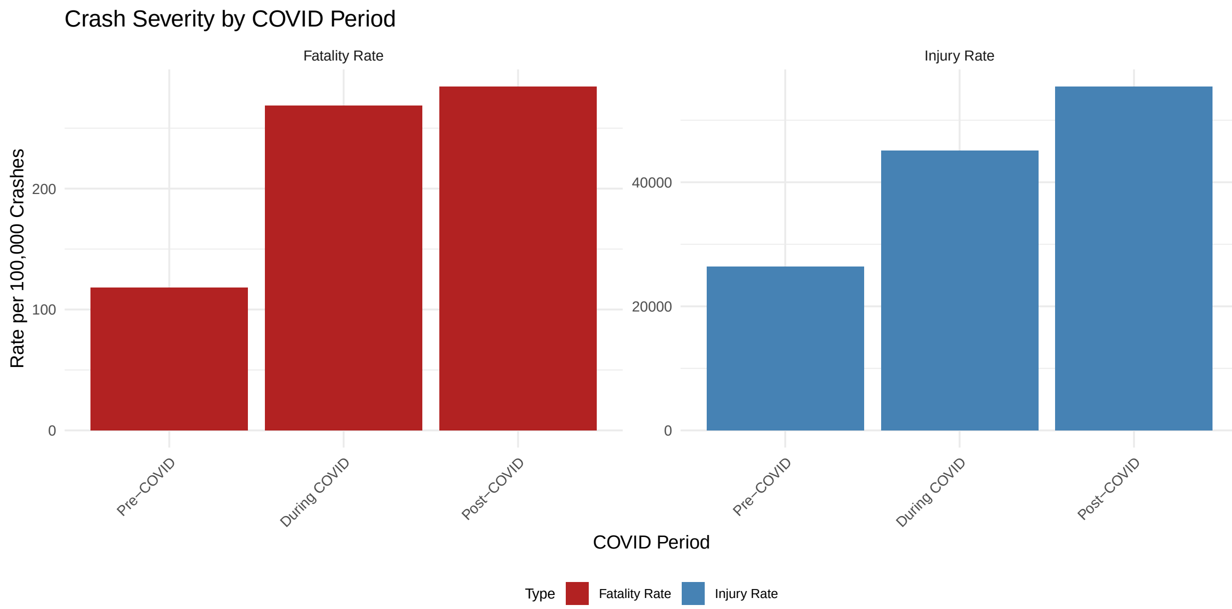
3.2.2 Hourly Crash Distribution by COVID Period



After examining overall daily and weekly crash patterns, we next examined how these hourly crash patterns evolved across the Pre-COVID, During-COVID, and Post-COVID periods. The proportional distribution of crashes by hour shows that daily patterns remained broadly consistent across the three COVID periods, but the intensity of rush-hour peaks changed. Before the pandemic, collisions followed a clear bimodal pattern with pronounced morning (around 8AM) and evening (5–6PM) peaks. During COVID, these peaks flattened and shifted slightly later in the day, reflecting reduced commuting and more flexible travel times. In the post-COVID period, the peaks partially recovered but stayed lower than pre-pandemic levels, suggesting that hybrid work and lasting behavioral changes continue to moderate traditional rush-hour peaks.

3.3 Did crashes become more severe?

We assessed whether crash severity changed during the pandemic by examining both the temporal patterns of crash severity and the direct impact of COVID-19 on fatality rates.



The left plot compares the rates of fatal and injury crashes by hour of day to assess whether crash severity changed during the pandemic. The injury rate follows a clear daily pattern, dipping in the early morning hours and rising sharply during daytime traffic peaks. Fatality rates remain much lower overall but show a slight increase overnight, when traffic volumes are low and speeds are higher. As expected, fatalities are far less common than injuries, with the fatality rate remaining close to zero compared to the much higher injury rate.

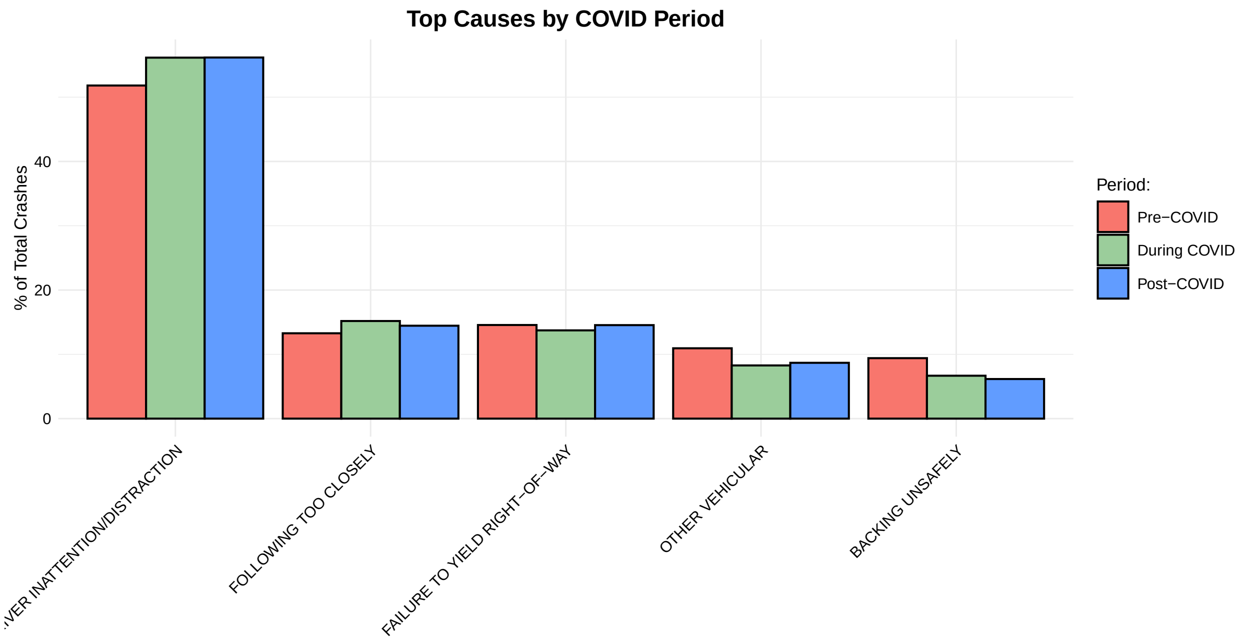
These patterns suggest that while the total number of crashes declined during the pandemic, the relative risk of severe outcomes did not decrease proportionally. Nighttime crashes continued to be far deadlier, reflecting persistent high-risk driving behaviors such as speeding and impaired driving.

The right barplot shows how crash severity shifted across COVID periods. During the pandemic, the fatality rate per 100,000 crashes increased, while the injury rate declined, indicating that although fewer collisions occurred, they were more severe. This pattern suggests that reduced traffic volume led to higher driving speeds and riskier behavior. In the post-COVID period, injury rates partially rebounded, but fatality rates remained elevated, implying that some high-risk driving habits persisted even after traffic returned to normal levels. Overall, the plot highlights a lasting change in crash severity linked to pandemic-era driving behavior.

3.4 What causes crashes? How did contributing factors change?

3.4.1 Causes of Collisions

To identify the most common causes of collisions, we aggregated all recorded contributing factors across vehicles. The bar chart below shows the five most frequent causes citywide.

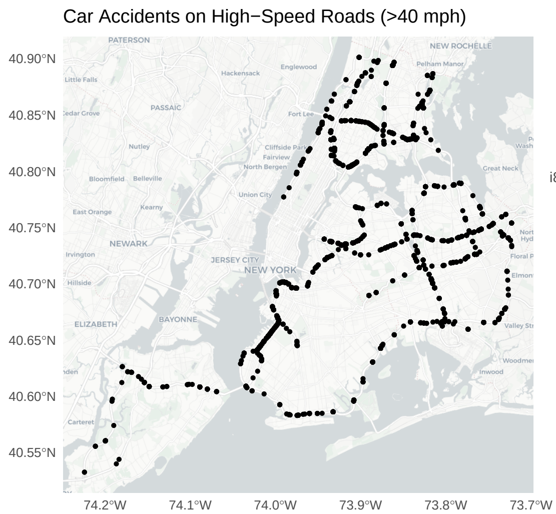


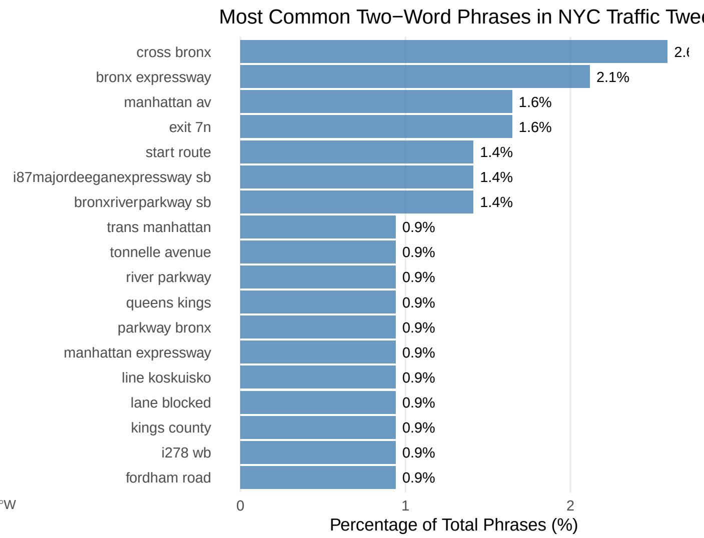
Driver inattention or distraction stands out as the leading cause, responsible for more than three times as many crashes as the next factor. Failure to yield, following too closely, and unsafe passing follow far behind, showing that most collisions stem from everyday driving mistakes rather than external conditions.

4 Text Mining: Twitter Traffic Reports

To complement our analysis of official crash data, we examined how traffic incidents are discussed on social media. Starting with 51,000 traffic-related tweets from 2018, we filtered to NYC-specific mentions and isolated vehicle crashes (excluding transit incidents), resulting in a focused dataset of over 700 crash-related tweets. While this data predates COVID-19, it provides insight into how real-time crash information is communicated and which locations generate the most traffic alerts.

The word cloud reveals that Twitter users predominantly describe crashes using geographic identifiers rather than causal factors. The most prominent phrases, "cross bronx," "bronx expressway," "manhattan av," and "exit 7n", emphasize where incidents occur. This pattern reflects Twitter's primary function as a real-time navigation tool, where drivers need to know which routes to avoid rather than why crashes happened.





The map on the left shows the crashes on high-speed roads (over 40 mph), highlighting major expressways and highways that dominate crash-related tweets. The bar chart on the right quantifies the most frequent location phrases, with "cross bronx" appearing in 2.6% of all crash-related bigrams—nearly double the frequency of most other mentions. This concentration reflects both the Cross Bronx Expressway's notoriously high crash rate and its critical role as a navigation landmark that affects thousands of commuters.

Other frequently mentioned locations include the Bronx River Parkway, Major Deegan Expressway (I-87), Manhattan Avenue, and specific highway exits. The prevalence of these major roadways in social media traffic reports aligns with the spatial distribution shown in the map, where crashes cluster along the city's primary highway corridors. This analysis demonstrates how social media serves as a complementary data source to official crash reports, capturing the real-time impact of incidents on traffic flow and providing crowd-sourced navigation alerts.

5 Statistical Modeling & Interpretation

5.1 Part 1: Baseline Risk Factors for Fatal Crashes

To investigate how the dynamics of fatal motor vehicle collisions in New York City shifted following the onset of the COVID-19 pandemic, we developed an interaction-based logistic regression model. The outcome of interest remains binary: whether a crash was fatal (at least one person killed) or non-fatal. However, unlike a standard baseline model, this approach allows us to explicitly test whether the relationship between risky behaviors and fatality rates changed significantly between the Pre-COVID and Post-COVID periods.

In simple terms, this model performs a difference-in-differences analysis. It compares crash patterns across thousands of incidents to answer two distinct questions:

1. Baseline Risk: Which factors (e.g., motorcycles, speeding) are inherently dangerous? dangerous?
2. The "COVID Effect": Did these specific factors become significantly more lethal after the pandemic began, even after accounting for general background trends?

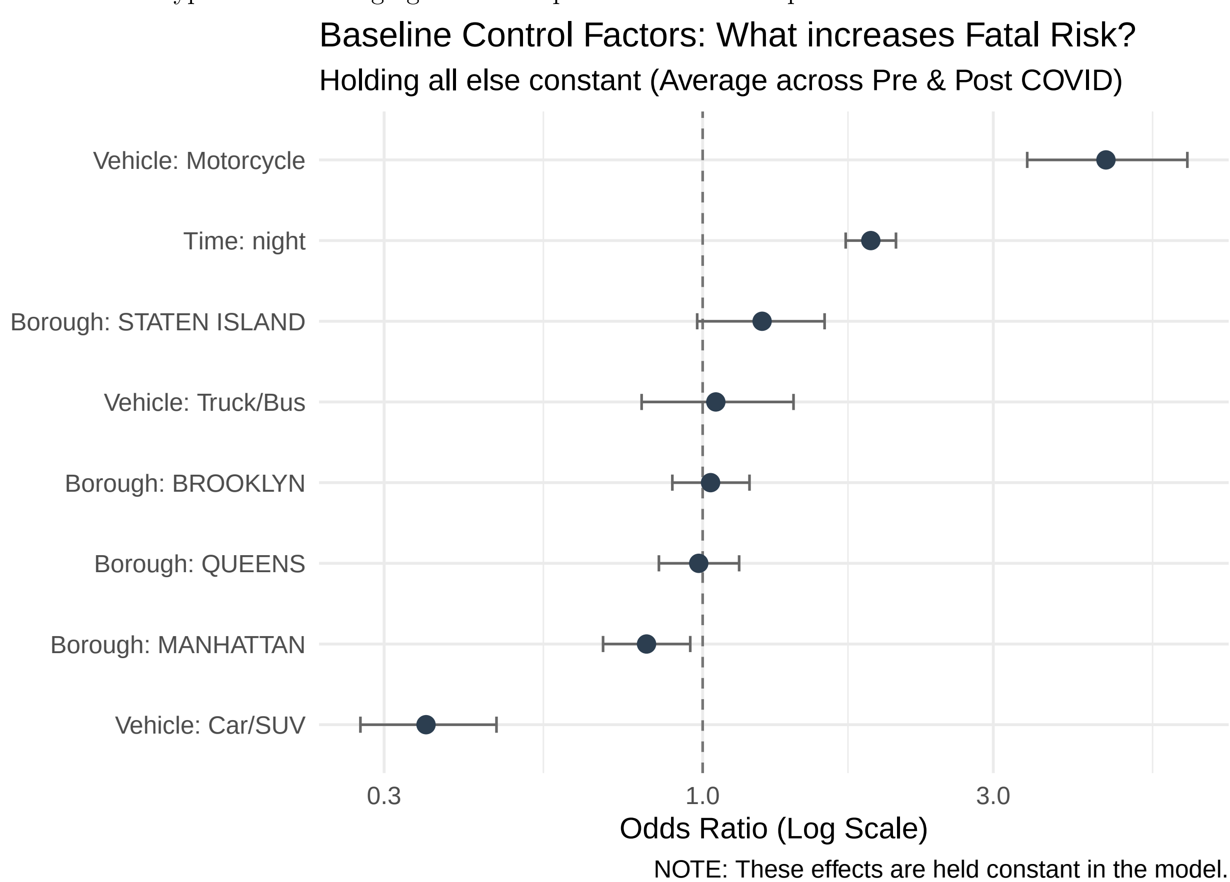
To ensure statistical robustness, we aggregated over 50 granular police-reported contributing factors into seven high-level behavioral categories (such as Impairment, Speeding/Aggression, and Distraction). By interacting these categories with the time period and using "Other/Unknown" crashes as a control group, the model isolates distinct behavioral shifts from general systemic changes. We also controlled for borough, time of day, and vehicle type to ensure that observed spikes in fatality risk were due to driver behavior rather than location or vehicle differences.

To interpret the results of the logistic regression, we rely on the Odds Ratio. This metric functions as a risk multiplier, quantifying how strongly a specific factor is associated with a fatal outcome compared to the baseline scenario.

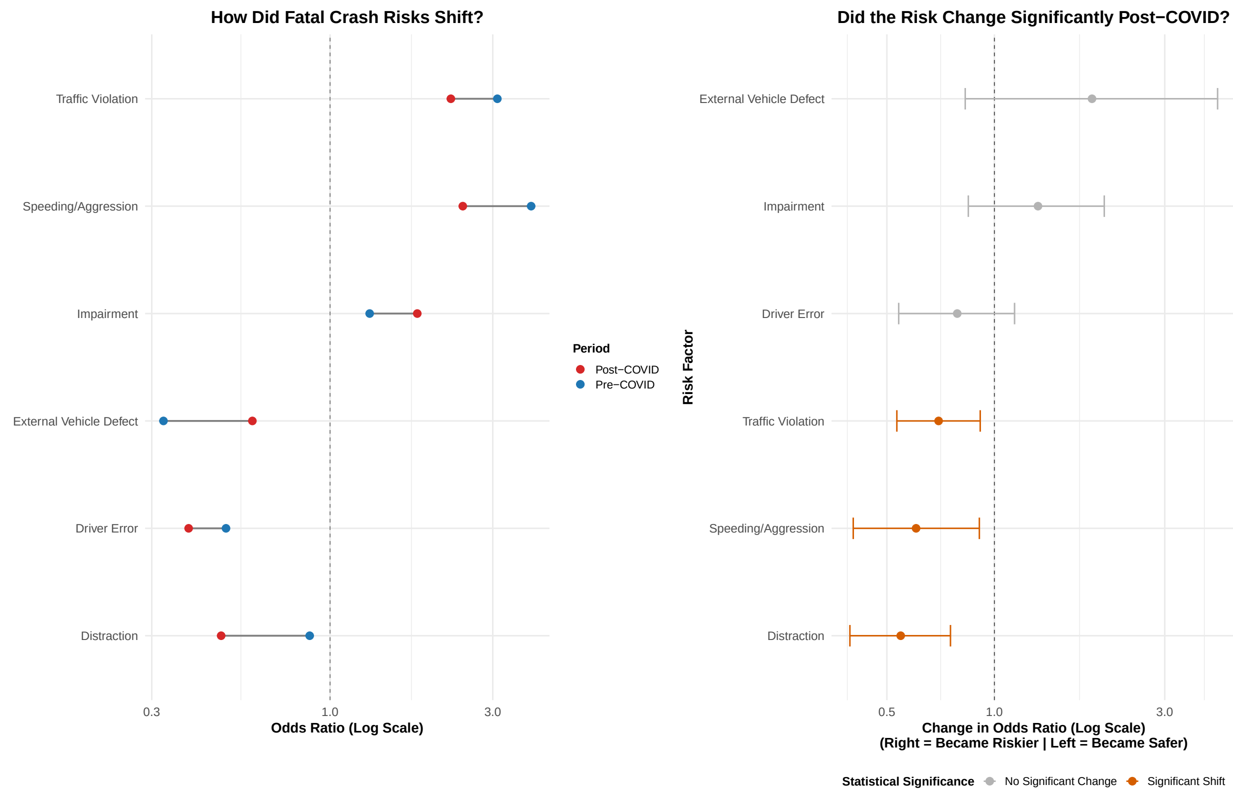
The values are interpreted as follows:

- An Odds Ratio of 1.0 indicates no difference in risk. The factor has no impact on the likelihood of a fatality compared to the baseline.
- An Odds Ratio greater than 1.0 indicates an increased risk. For example, a value of 3.0 suggests that the presence of this factor makes a fatal outcome three times more likely than if the factor were absent.
- An Odds Ratio less than 1.0 indicates a decreased risk. This suggests the factor has a protective effect, making a fatal outcome less likely.

While the primary focus of this analysis is the shift in driver behavior, it is crucial to isolate these effects from the inherent physical risks of different crash scenarios. To achieve this, our model controls for Borough, Vehicle Type, and Time of Day as fixed effects. This adjustment accounts for known structural confounders (such as the heightened lethality of motorcycle collisions or the reduced visibility of nighttime driving) which exist independently of the pandemic. By holding these variables constant, we ensure that the significant changes observed in the behavioral categories represent a genuine shift in the lethality of driver conduct, rather than a byproduct of changing traffic composition or location patterns.



This plot illustrates the fixed effects of the control variables (Vehicle Type, Time of Day, and Borough), holding all behavioral factors constant. These results validate the model's reliability, as they align with established traffic safety principles. The most significant predictor of fatality is vehicle type: incidents involving motorcycles show a drastically elevated risk (Odds Ratio ¿ 3.0) compared to the baseline, confirming that motorcyclists remain the most vulnerable road users regardless of other conditions. Time of day is also a significant factor, with nighttime crashes showing nearly double the odds of a fatal outcome compared to daytime incidents, likely due to reduced visibility and higher travel speeds. Geographically, Manhattan is associated with the lowest fatality risk, likely a result of congestion and lower average speeds. On the other hand, Staten Island, with its more suburban, highway-centric infrastructure, presents a higher baseline risk.



This figure presents a comprehensive view of how driver behaviors shifted during the pandemic. The left panel illustrates the absolute risk levels, identifying Speeding/Aggression and Traffic Violations as the most lethal factors in both periods (Odds Ratios ¿ 2.0).

However, the right panel reveals crucial nuances in how these risks changed. Notably, Distraction and Traffic Violations showed a statistically significant decrease in lethality relative to the baseline trends (Interaction OR ¡ 1.0, $p < 0.05$). In contrast, while Impairment appeared to become more dangerous in the raw data (Left Panel, Red dot ¿ Blue dot), the interaction model (Right Panel, Gray bar) confirms that this shift was not statistically distinguishable from chance ($p > 0.05$). This suggests that while distracted driving became less predictive of fatalities, the extreme danger of impaired driving remained a stable constant throughout the pandemic.

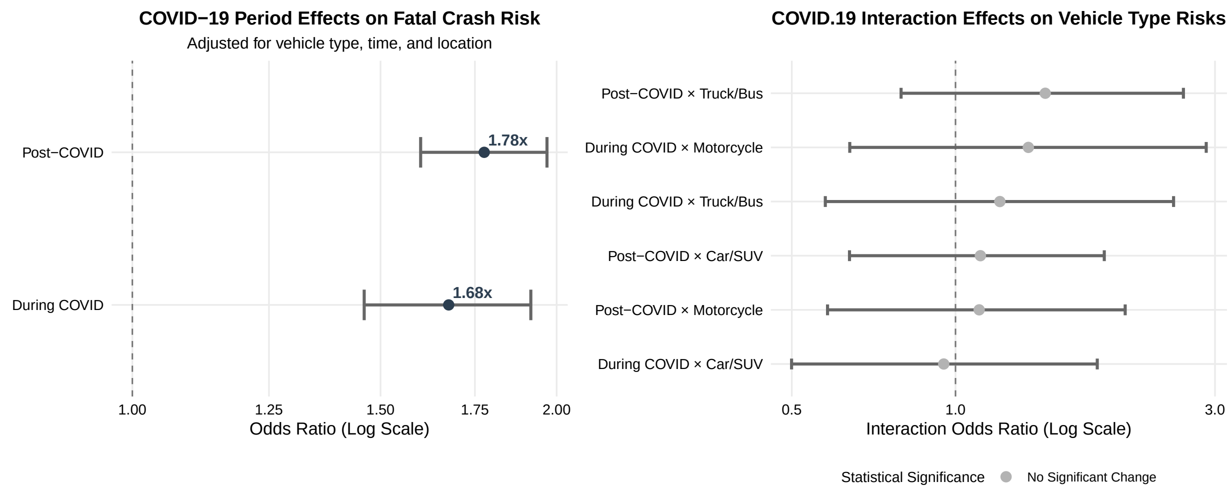
5.1.1 Interpretation

In summary, this analysis utilized an interaction-based logistic regression to disentangle the specific effects of the COVID-19 pandemic from established structural risk factors in NYC collisions. By controlling for fixed effects such as vehicle type and borough, we confirmed that physical determinants of fatality—most notably motorcycle involvement and nighttime driving—remained consistent predictors of mortality regardless of the pandemic.

However, the interaction model revealed that the "COVID Effect" was not a uniform amplifier of risk. While Speeding/Aggression and Impairment remained persistently high-risk behaviors with no statistically significant deviation from their baseline lethality, Distraction and Traffic Violations exhibited a significant relative decline in their association with fatal outcomes. These findings underscore that the post-pandemic traffic landscape was not defined by a blanket increase in danger across all categories, but rather by a specific restructuring of how certain driver behaviors translate into fatal outcomes.

5.2 Part 2: COVID-19's Impact on Crash Severity

Building on our baseline risk analysis, we developed statistical models to quantify how COVID-19 specifically altered crash severity patterns in NYC. This analysis tests whether the pandemic period itself was independently associated with changes in fatality risk, controlling for established risk factors.



5.2.1 Interpretation

After controlling for borough, vehicle type, and time of day, crashes during the COVID period were 1.68 times more likely to be fatal (95% CI: [1.45, 1.91]), and post-COVID crashes remained 1.78 times more likely (95% CI: [1.61, 1.98]) compared with the pre-pandemic baseline. The persistence of this elevated risk indicates that the pandemic produced lasting structural changes in crash dynamics rather than temporary shifts in traffic composition.

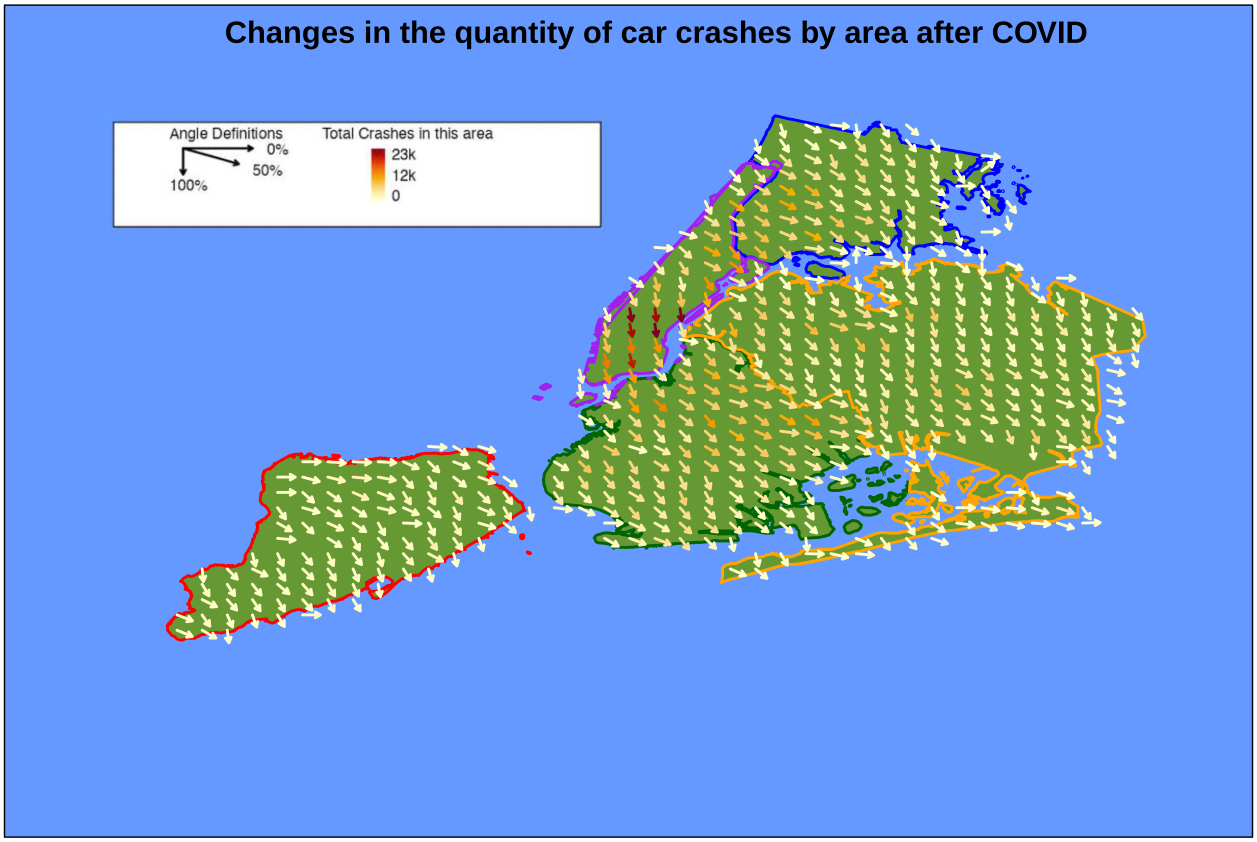
The statistical models confirm that COVID-19 had a significant and substantial effect on crash severity. In the unadjusted model, crashes during COVID were 1.91 times more likely to be fatal (95% CI: [1.67, 2.19]), and post-COVID crashes were 2.12 times more likely (95% CI: [1.92, 2.34]) than before the pandemic. Even after adjustment, the effect persisted, reinforcing that the increase in fatality risk cannot be explained solely by differences in exposure or vehicle mix.

The interaction analysis (left plot) shows that while overall severity rose across all vehicle types, the relative risk hierarchy remained stable. No vehicle category exhibited statistically significant differential effects at the 0.05 level, although motorcycle crashes trended toward higher risk during COVID ($p = 0.097$) and truck/bus crashes trended upward post-COVID ($p = 0.100$). These patterns suggest that the pandemic amplified crash severity uniformly rather than disproportionately affecting specific vehicle classes.

6 Killer Plot

6.1 Static Killer Plot

Our static killer plot compares the quantity of collisions Pre-COVID to the quantity of collisions Post-COVID by area. For each we take a three year range, separating that range enough from the actual COVID time period to ensure that we are not measuring data from during the pandemic. We then divided up the map of NYC into a grid, calculating the change in collisions within each grid. We can see that all throughout NYC, the amount of crashes decreased. However, the amount that it decreased by was area-dependent. In Manhattan, we can see the largest drop off by far, with the least effected borough being Staten Island. We see similar trends in Queens, the Bronx, and in Brooklyn. We also color coded the map so that there is an emphasis placed on areas that had a higher amount of accidents overall, with the most being in Manhattan, and the least being in Staten Island. The amount of accidents seems to correlate with the change in accidents.



6.2 Dynamic Killer Plot

To further analyze these trends, we created two separate interactive models, available through the killer_plot.R file in our project directory. Running this file launches a Shiny application that allows dynamic exploration of crash data.

The first model shows us the relative quantity of accidents in a given year by area. Users can select a year, and the grid will populate with dots scaled by the quantity of accidents. This allows us to clearly see the difference in the accident distribution pre and post COVID In the years before COVID, the dots are far larger in Manhattan, as well as some areas of Brooklyn and the Bronx. However, if you change the year to a year that is Post-COVID, such as 2022, this distribution is far more even, with not much of a difference between Manhattan, Brooklyn, and the Bronx.

Users can switch the tab to see the other interactive model, which shows the accident trends in a given range of years. Users can select a range of years, such as 2015-2018, and a linear regression will be performed in each segment of the map. An arrow then displays the angle of that regression in each area. We can see in this Pre-COVID selection that Manhattan was on a downward trend, while all the other boroughs were on an upward trend, with Staten Island being the most static, and the Bronx having a pretty significant upward trend in many areas. We can then switch the range to be Post-COVID, selecting 2020-2024. Now, there is a significant downward trend in the Bronx, Brooklyn, and Queens, but most significantly in the Bronx. Manhattan is still on a downward trend, but less so than in the previous year selection, and Staten Island is almost completely static, actually increasing in some areas while decreasing in others. We can see that not only did the distribution of accidents change after COVID, but the trends in each area changed as well.

7 Conclusion

7.1 Key Takeaways

1. Did collision rates return to pre-pandemic levels or establish a "new normal"?: A clear new normal was established. While collision volumes dropped sharply in March 2020, they stabilized at consistently lower levels rather than returning to pre-pandemic baselines. Brooklyn and Queens showed the largest sustained declines, while Manhattan rebounded more quickly. We used a piecewise linear regression that confirmed distinct pre- and post-COVID trends, which was caused by hybrid work and altered commuting patterns creating changes in traffic volume and timing post-COVID.

2. Did crashes become more or less deadly during reduced traffic?: After the pandemic we saw fewer crashes, but deadlier outcomes. Controlling for vehicle type, location, and time of day, crashes during COVID were 68% more likely to be fatal, and post-COVID crashes remain 78% more deadly than pre-pandemic levels. Empty roads probably enabled higher speeds and riskier behaviors that persisted even as traffic returned. Injury rates declined while fatality rates spiked.

3. Were impacts uniform across NYC's five boroughs?: Manhattan, with its congested streets and lower speeds, maintained the lowest fatality risk and fastest crash-rate recovery. Brooklyn and Queens experienced the largest declines and slowest recoveries, consistent with commuter-heavy traffic patterns. Staten Island's suburban, highway-centric infrastructure showed higher baseline fatality risk throughout all periods. The Bronx demonstrated moderate impacts with gradual recovery.

4. How did vehicle characteristics and driver behaviors contribute to collision severity across periods?: Motorcycles remained the highest-risk vehicle type ($3\times$ baseline fatality odds) across all periods, with nighttime driving nearly doubling fatal crash risk. Driver inattention/distraction dominated as the leading cause in all periods. However, interaction modeling showed nuanced changes: while speed-ing/aggression and impairment maintained consistently high lethality, distraction and traffic violations showed statistically significant decreases in their association with fatal outcomes post-COVID.

7.2 Potential Future Work

This analysis could be extended in several directions. Future research could examine how behavioral changes during COVID influenced risk-taking on empty roads, assess whether Vision Zero policies remained effective during shifting traffic patterns, or develop predictive models incorporating real-time traffic and weather data. Comparing NYC's experience to other major cities would help identify which patterns are universal versus city-specific. Continued monitoring will also be valuable to see whether these post-COVID trends stabilize or continue evolving as work-from-home policies mature.